# DROP
# Version 2.30
*Updated March 31st, 2015*

## 1 Overview

NASDAQ accepts limit orders from subscribers and executes matching orders when possible. Non-matching orders may be added to the NASDAQ Book, a database of available limit orders, where they wait to be matched in price-time priority.
DROP is a protocol that delivers real-time information about activity that takes place on the NASDAQ.

Each DROP account is configured to transmit information concerning orders entered by one or more NASDAQ subscriber firms. DROP is typically used by clearing firms to track the activity of their correspondents, or by larger firms to monitor the activity of multiple NASDAQ access points for risk management purposes.

Each DROP host can be configured to send a message anytime an order is entered, canceled, executed, or broken – or any combination of these events.

DROP does not provide the ability to enter orders into NASDAQ.

### *1.1 Architecture*

The DROP protocol is composed of logical messages passed between the DROP host and the client application. Each message type has a fixed message length.

All messages sent from the DROP host to the client are assumed to be sequenced, and their delivery must be managed by some lower level protocol. The SoupTCP or SoupBinTCP (available separately) are the typical lower level protocols used to guarantee the delivery and sequencing of DROP messages sent from the host to the client.

### *1.2 Recovering From Broken Connections*

In the case where a client looses the connection to the DROP host and wishes to reconnect without having to re-read though all the messages it has already received, there is an optional line number parameter that can be added to the end of the password line when logging in. The format of this login line is…

password[,line number]

where “password” is the assigned client password and line number is the optional line number the client would like the host to begin transmission with. The login line is always terminated with a CR/LF pair or just an LF. If the optional line number is not specified, the DROP host always begins transmission with the first message for the current day (line #1).

By counting incoming lines, the client can re-connect and request the precise next expected line number and prevent any redundant messages.

## 1.3 Data Types

Numeric fields are a string of ASCII coded digits, right justified and space filled on the left.

Alpha fields are left justified and padded on the right with spaces.

Prices are given in decimal format with 6 whole number places followed by a decimal point and 4 decimal digits. The whole number portion is padded on the left with spaces; the decimal portion is padded on the right with zeros.

Timestamps are numeric given in seconds past midnight Eastern Time.

## 1.4 Fault Redundancy

Multiple DROP hosts can be configured to send information on an identical set of events and matching firms and ports, making it possible to create mirrored DROP hosts for purposes of fault redundancy.

For maximum redundancy, the mirrored machines should be located at geographically diverse data centers with communications carrier access diversity. The two lines could also terminate at different subscriber locations on distinct computing platforms.

## 1.5 Service Bureau Configuration

A single DROP host can deliver information for one or more firms, allowing a service bureau configuration. In this case, the DROP account must be authorized by each desired firm using a DROP Port Authorization Form.

## 1.6 Trade Message Line Format

Once logged in, the client will receive a series of message lines from the host in real time. Each message line is fixed format, comma delimited, and CR/LF terminated ASCII text.

| Name | Offset | Len | Type | Sample | Notes |
|---|---|---|---|---|---|
| Time Stamp | 0 | 9 | Timestamp | 34293.1 | The time the event occurred on INET to the nearest millisecond. |
| Type | 10 | 1 | Type | 'E' | "A"=New order accepted<br>"E"=Existing order executed<br>"X"=Existing order canceled<br>"B"=Previous execution broken<br>"U"=Existing order replaced<br>“M”=Existing order modified<br>“Y”=Existing order cancelled (AIQ)<br>“C”=Trade correction |
| Source | 12 | 6 | Alphanum | ABCD01 | The source of the order. Typically the account of the OUCH port used to enter the order, but can also have the special values of “$PHON ” for orders received via NASDAQ’s phone desk. |

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| Order Token | 19 | 15 | Alphanum | (arbitrary) | The free form Token field as specified by the order entry firm when the order was entered into NASDAQ. The field is defined as user(4)+comma(1)+token(10) |
| Replaced Token | 35 | 10 | Alphanum | (arbitrary) | The free form Token field as specified by the order entry firm when the order was entered into NASDAQ. Only used when we send a replaced message. |
| Buy/Sell | 46 | 1 | Alpha | B | The side of the trade executed. B=Bought, S=Sold, T=Sold Short, E=Sold Short Exempt |
| Shares | 48 | 6 | Numeric | 10000 | For a new order accept, the total number of shares entered.<br><br>For an existing order cancel, the incremental number of shares canceled. Note that an order can be partially canceled and still have open shares.<br><br>For an existing order execute, the incremental number of shares executed in this trade. Note that a single order can result in multiple executions.<br><br>For a broken execution, the number of shares in the previously transmitted execution. |
| Stock | 55 | 8 | Alpha | INTC | The stock symbol |
| Price | 64 | 11 | Numeric with 4 decimal places | 12.875 | For a new order accepted, the limit price of the order.<br><br>For an execution, the execution price.<br><br>For a cancel, the limit price of the open order. |
| Firm | 76 | 4 | Alpha | BIGJ | The order entry firm. |
| Reference | 81 | 12 | Numeric | 836455 | The order unique reference number assigned by NASDAQ to this order. |

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| Match/TIF | 94 | 12 | Numeric | 122853 | For executions and breaks, The match number assigned by NASDAQ to this trade. Each match consists of an execution between a buy order and a sell order.<br><br>For other messages, this field will represent Time in Force (TIF). |
| Capacity | 107 | 1 | Alpha | A | The capacity as specified by the order entry firm.<br>A=Agency<br>P=Principal<br>R=Riskless |
| Liquidity Code / Cancel reason | 109 | 1 | Alpha | R | For execution messages, this field will represent the liquidity code value:<br>A=Added<br>R=Removed<br>X=Routed<br>D=DOT<br>F = Opening Trade (on NYSE)<br>G = On-Close order (on NYSE)<br>O=Opening Cross<br>M=Opening Cross (imbalance-only)<br>C=Closing Cross<br>L=Closing Cross (imbalance-only)<br>H=Halt/IPO Cross<br>K=Halt Cross<br><br>J = Non-displayed adding liquidity<br>Y = Re-Routed by NYSE<br>S = Odd Lot Execution (on NYSE)<br>U = Added Liquidity (On NYSE)<br>B = Routed to BX<br>E = NYSE Other<br>P = Routed to PSX<br>T = Opening Trade (on ARCA)<br>W = Added post-only (not currently available)<br>Z = On-Close order (on ARCA)<br>m = Removed liquidity at a midpoint<br>k = Added liquidity via a midpoint order0 = Supplemental order execution<br>7 = Displayed, liquidity-adding |

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| | | | | | order improves the NBBO<br>8 = Displayed, liquidity-adding order sets the QBBO while joining the NBBO<br>d = Retail designated execution that removed liquidity (not currently available)<br>e = Retail designated execution that added displayed liquidity<br>f = Retail designated execution that added non-displayed liquidity (not currently available)<br>j = RPI order that provides liquidity<br>r = Retail order that removes RPI liquidity<br>t = Retail order that removes price improving non-displayed liquidity other than RPI liquidity<br>4 = Added displayed liquidity in a Select Symbol<br>5 = Added non-displayed liquidity in a Select Symbol<br>6 = Liquidity Removing Order in a Select Symbol<br>g = Added non-displayed midpoint liquidity in a Select Symbol<br><br>For cancel messages, this field will represent the cancel reason value:<br>U = User cancel<br>I = IOC cancel<br>T = Timeout<br>S = Supervisory cancel<br>D = Regulatory cancel<br>Q = Self Match Prevention<br>Z = System cancel<br>C = Cross cancel |
| Clearing Code | 111 | 1 | Alpha | A | The clearing path this trade will take. Q=QSR |
| ISO Flag | 113 | 1 | Alpha | Y | Y = eligible<br>N = not eligible |
| Display | 115 | 1 | Alpha | Y | Accepted order display indicator |
| BBO Weighting Indicator | 117 | 1 | Numeric | 1 | 0 = 0-0.2%<br>1 = 0.2%-1%<br>2 = 1%-2%<br>3 = greater than 2% |

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| Reference Price | 118 | 11 | Numeric with 4 decimal places | 12.8751 | The reference price associated with the execution (may be 0 if not applicable) |
| Reference Price Type | 129 | 1 | Alpha | I | Valid values:<br><br><blank> = not applicable<br>I = Intraday Indicative Value |

# 2 Support

If you have any questions or comment about this specification, just E-mail to tradingservices@nasdaqomx.com. We also welcome any suggestions for new features or improvements.

# 3 Revision History

| Revision # | Date | Change |
|---|---|---|
| 2.3 | 04/25/2012 | Derived Version 2.3 from Version 2.2<br>Added ISO Flag field<br>Added Display field<br>Added BBO weighting indicator field<br>Added Support for Order Modified |
| 2.3 | 07/18/2012 | Added cancel reason to the liquidity flag field for cancelled order messages |
| 2.3 | 08/28/2012 | Added message type value “Y” |
| 2.3 | 10/23/2012 | Added liquidity flag values “7”, “8” |
| 2.3 | 01/10/2013 | Added liquidity flag values “d”, “e”, “f” |
| 2.3 | 01/28/2013 | Added liquidity flag values “6” |
| 2.3 | 06/10/2014 | Updated the formatting of the document to make more standardized across all specs |
| 2.3 | 01/15/2015 | Added Liquidity flag values “j”, “r”, “t”, “4”, “5”, “g” |
| 2.3 | 03/31/2015 | Drafting NextShares version of DROP<br>Added reference price<br>Added Reference Price Type |
| 2.3 | 07/14/2015 | Added message type value “C” |